Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

(Signature and Date in Handwritten)

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Session	**2026-27 (Odd)**	**Course Name**	**LAB: Optimization technique**
Semester	**7th**	**Course Code**	**23ADS1708**
Roll No	**39**	**Name of Student**	Hrutik Gujarkar

Practical Number	**5**
Course Outcome	CO1: Understand unconstrained and constrained optimization techniques. CO2: Apply non-linear optimization techniques to solve engineering problems. CO3: Analyze optimization problems using operations research techniques. CO4: Evaluate transportation and transshipment optimization problems.
Aim	To solve inequality-constrained optimization problems using the Kuhn-Tucker (KKT) Method.
Problem Definition	To minimize the objective function $f(x, y) = (x - 3)^2 + (y - 2)^2$ subject to the inequality constraints $x + y \leq 4$ and $x \geq 0$ using the Kuhn-Tucker (KKT) conditions, and to determine the optimum point and the optimum value of the objective function.
Theory (100 words)	The Kuhn-Tucker (KKT) method extends the Lagrange multiplier method to problems with inequality constraints. For minimizing $f(x)$ subject to $g_i(x) \leq 0$, the Lagrangian is $L(x, \mu) = f(x) + \Sigma\ \mu_i\ g_i(x)$. A point is a candidate optimum if it satisfies four KKT conditions: (1) stationarity, $\nabla L = 0$; (2) primal feasibility, $g_i(x) \leq 0$; (3) dual feasibility, $\mu_i \geq 0$; and (4) complementary slackness, $\mu_i\ g_i(x) = 0$. Complementary slackness means that every constraint is either active ($g_i = 0$) or has a zero multiplier. For a convex problem, a KKT point is the global minimum.

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<table>
<tr><td></td><td>Step-by-Step Method

1. Identify the objective function f(x, y) and write every constraint in the form $g_i(x, y) \leq 0$.

2. Form the Lagrangian: $L = f + \mu_1 g_1 + \mu_2 g_2$.

3. Write the stationarity conditions $\partial L/\partial x = 0$ and $\partial L/\partial y = 0$.

4. Consider every combination of active constraints ($g_i = 0$) and inactive constraints ($\mu_i = 0$).

5. For each case, solve the resulting equations for x, y, μ_1 and μ_2.

6. Check primal feasibility ($g_i \leq 0$) and dual feasibility ($\mu_i \geq 0$) for the solution of each case.

7. The case that satisfies all KKT conditions gives the optimum point; substitute it in f(x, y) to obtain the optimum value.</td></tr>
<tr><td rowspan="2">Procedure and Execution (100 Words)</td><td>Algorithm:
1. Define the objective function.
2. Define the inequality constraints in the form $g(x, y) \leq 0$.
3. Construct the Lagrangian function.
4. Compute the stationarity conditions $\partial L/\partial x$ and $\partial L/\partial y$.
5. For every combination of active/inactive constraints, solve the system of equations.
6. Check primal feasibility and dual feasibility ($\mu \geq 0$).
7. Select the case that satisfies all KKT conditions.
8. Calculate the objective function value at the optimum point.
9. Display the optimal solution and objective function value.</td></tr>
<tr><td>Code:
<pre>
import sympy as sp
import itertools

define variables
x, y = sp.symbols("x y", real=True)
mu1, mu2 = sp.symbols("mu1 mu2", real=True)

Objective function (minimize)
f = (x - 3)**2 + (y - 2)**2

Inequality constraints in the form g(x, y) <= 0
g1 = x + y - 4   # x + y <= 4
g2 = -x   # x >= 0
g = [g1, g2]
mu = [mu1, mu2]

Lagrangian function
</pre></td></tr>
</table>

```
L = f + mu1*g1 + mu2*g2
print("Lagrangian Function:")
print(L)

# Stationarity conditions
dLdx = sp.diff(L, x)
dLdy = sp.diff(L, y)
print("\nStationarity Conditions:")
print("dL/dx =", dLdx)
print("dL/dy =", dLdy)

# Check every combination of active / inactive constraints
print("\nCase Analysis:")
best = None
cases = itertools.product([0, 1], repeat=2)
for case_no, active in enumerate(cases, start=1):
    eqs = [dLdx, dLdy]
    for i in range(2):
        if active[i]:
            eqs.append(g[i])      # active: g_i = 0
        else:
            eqs.append(mu[i])     # inactive: mu_i = 0
    sol = sp.solve(eqs, [x, y, mu1, mu2], dict=True)
    print(f"\nCase {case_no}: g1 active = {bool(active[0])}, "
          f"g2 active = {bool(active[1])}")
    if not sol:
        print("  No solution")
        continue
    s = sol[0]
    xv, yv = s[x], s[y]
    m1, m2 = s.get(mu1, 0), s.get(mu2, 0)
    feasible = all(gi.subs({x: xv, y: yv}) <= 0 for gi in g)
    dual_ok = (m1 >= 0) and (m2 >= 0)
    print(f"  x = {xv}, y = {yv}, mu1 = {m1}, mu2 = {m2}")
    print(f"  Primal feasible: {feasible}, "
          f"Dual feasible (mu >= 0): {dual_ok}")
    if feasible and dual_ok:
        print("  --> KKT conditions satisfied")
        best = (xv, yv, m1, m2)

# Optimal solution
xv, yv, m1, m2 = best
print("\nOptimal Solution:")
print("x =", xv)
print("y =", yv)
print("mu1 =", m1)
print("mu2 =", m2)
print("\nMinimum Value of Objective Function:")
print(f.subs({x: xv, y: yv}))
```

Output:

```
Lagrangian Function:
mu1*(x + y - 4) - mu2*x + (x - 3)**2 + (y - 2)**2

Stationarity Conditions:
dL/dx = mu1 - mu2 + 2*x - 6
D
```

	`L/dy = mu1 + 2*y - 4`
	
	`Case Analysis:`
	
	`Case 1: g1 active = False, g2 active = False`
	`  x = 3, y = 2, mu1 = 0, mu2 = 0`
	`  Primal feasible: False, Dual feasible (mu >= 0): True`
	
	`Case 2: g1 active = False, g2 active = True`
	`  x = 0, y = 2, mu1 = 0, mu2 = -6`
	`  Primal feasible: True, Dual feasible (mu >= 0): False`
	
	`Case 3: g1 active = True, g2 active = False`
	`  x = 5/2, y = 3/2, mu1 = 1, mu2 = 0`
	`  Primal feasible: True, Dual feasible (mu >= 0): True`
	`  --> KKT conditions satisfied`
	
	`Case 4: g1 active = True, g2 active = True`
	`  x = 0, y = 4, mu1 = -4, mu2 = -10`
	`  Primal feasible: True, Dual feasible (mu >= 0): False`
	
	`Optimal Solution:`
	`x = 5/2`
	`y = 3/2`
	`mu1 = 1`
	`mu2 = 0`
	
	`Minimum Value of Objective Function:`
	`1/2`

Output Analysis	The program constructs the Lagrangian and checks all four active/inactive cases. Case 1 gives the unconstrained minimum (3, 2), which violates $x + y \leq 4$. Cases 2 and 4 give negative multipliers, so dual feasibility fails. Only Case 3, where $x + y \leq 4$ is active and $x \geq 0$ is inactive, satisfies all KKT conditions with $x = 5/2$, $y = 3/2$, $\mu_1 = 1$ and $\mu_2 = 0$. The objective function value is $f(2.5, 1.5) = 0.5$. Since f is convex and the constraints are linear, this is the global minimum.
Link of student Github profile where lab assignment has been uploaded https://github.com/HrutikGujarkar/Optimizaton-techniques-Practicals	

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Conclusion	The inequality-constrained optimization problem was successfully solved using the **Kuhn-Tucker (KKT) Method**. The optimal point is (2.5, 1.5) with multipliers $\mu_1 = 1$ and $\mu_2 = 0$, and the minimum value of the objective function is **0.5**.
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